



BEFORE

Little is known about the development of human speech and conscious thought. Physical evidence can yield some clues.

LOOKING AT THE EVIDENCE

Internal casts of human skulls (endocasts) reveal the relatively small brains of *Australopithecus* << 16–19 as apelike and incapable of speech.

A BRAIN FIT FOR THE JOB

The **brain size** of our early ancestors **grew gradually over millions of years**, allowing increasing levels of sophistication in communication and culture. *Homo sapiens*' brain measures 97½ cu in (1,600 cm³), almost three times the size of that of *Homo habilis*, whose brain capacity was 36½ cu in (600 cm³).

NO TALKING

Homo habilis << 16–19, who lived from about 2.5 million years ago, is thought to have had very limited communication skills, possibly using a range of **signs and grunts** to foster cooperation between members of a group.

The Art of Communication

Speech and language were key developments in human history, perhaps even more so than toolmaking. They turned the simple signs and grunts of our ancestors into increasingly sophisticated communication. Archaeology and studies of human anatomy help to indicate when these important traits evolved.



Discovering speech

The hyoid bone is found in the neck and is required for speech to occur. Finds such as these fossilized pieces of 400,000-year-old *Homo heidelbergensis* hyoid bone from Atapuerca, Spain, help date the first human speech.

Our knowledge about when and how speech evolved remains a controversial area in the study of early human history. Articulate speech is an important threshold in human evolution because it opened up new vistas of cooperative behavior and the enrichment of human life. From archaeological evidence alone, it is difficult to know accurately when speech first developed. *Homo habilis* had a slightly more humanlike frontal lobe (where speech control is located) than earlier australopithecines. Other clues are found in the position of the larynx (voice box)—unlike all other mammals, the larynx of *Homo sapiens* is positioned low, permitting a wide variety of vocal

sounds. *Homo erectus*, from around 1.8 million years ago, was the first human with a lower larynx, and finds from Sima de los Huesos, in Atapuerca, Spain have shown that *Homo heidelbergensis* had developed a hyoid bone—a small, U-shaped bone that lies at the root of the tongue, between the larynx and pharynx—about 400,000 years ago. It was only about 300,000 years ago, however, that the base of the skull evolved, physically allowing fully articulate speech to develop.

The Neanderthal debate

Neanderthals may have had some capacity for speech and communication, and were apparently capable of